1) A load $Z_L = 50 + j10\ \Omega$ is connected to a generator $V_G = 10\angle 0°$ (V) with a 30.48-m, 50-ohm lossless transmission line. The generator's internal impedance is 20 Ω, the operating frequency is 10 MHz, and the wave velocity along the line is (2/3)c. (a) Find the line input voltage. (b) Find the input current. (c) Find the line VSWR. (d) Find the average power along the line.

$\lambda = 20\,m \Rightarrow \beta = 0.3142\ (rad/m) \Rightarrow \beta\ell = 9.5756\ (rad)$

$Z_{in} = 50\left[\dfrac{50 + j10 + j50\tan(\beta\ell)}{50 + (j50 - 10)\tan(\beta\ell)}\right] = 53.11 + j9.83\ (\Omega)$

a) $V_0 = 10\,\dfrac{53.11 + j9.83}{20 + 53.11 + j9.83} = 7.31 + j0.36\ (V)$

b) $I_0 = \dfrac{V_0}{Z_{in}} = 0.13 - j0.02 = 0.14\angle -7.66°\ (A)$

c) $|\Gamma| = 0.0995 \rightarrow VSWR = 1.22$

d) $|V_0^+| = \left|\frac{1}{2}(V_0 + Z_0 I_0)\right| = 6.912\ (V)$

$P_{ave} = \dfrac{|V_0^+|^2}{2Z_0}(1 - |\Gamma|^2) = 0.473\ (W)$

2) Consider a dielectric-filled metallic rectangular cavity of size $3 \times 4 \times 5$ (cm). (a) Find a suitable non-magnetic dielectric material to fill the cavity such that it has single-mode operation between 0-3 GHz. (b) For your design, find all excited modes below 6 GHz.

The dominant mode of this cavity is TE011

$$f_{011} = \frac{c}{2\sqrt{\varepsilon_r}}\sqrt{\left(\frac{0}{0.03}\right)^2 + \left(\frac{1}{0.04}\right)^2 + \left(\frac{1}{0.05}\right)^2} < 3 \times 10^9 \quad \rightarrow \quad 2.55 < \varepsilon_r < 3.77$$

Otherwise, the next mode (TE101) will be excited below 3GHz

$TE_{011} = 2.87\ GHz\ \ TE_{101} = 3.84\ GHz\ \ TM_{110} = 3.74\ GHz\ \ TE_{111} = TM_{111} = 4.14\ GHz \ldots..$

3) The plane wave described by the electric field below in free space, encounters a lossless non-magnetic dielectric medium of ε_r=5, occupying the space $y \le 0$.

$$\vec{E} = (10\,\vec{a}_y + 5\,\vec{a}_z)\,e^{j(2y-4z)}\quad (V/m)$$

a) Find the frequency and wavelength of the wave in each medium.
b) Find the angles of incidence and transmission and the coefficients of reflection and transmission.
c) Write the transmitted electric field expression.

$\vec{E} = (10\vec{a}_y + 5\vec{a}_z)e^{j(2y-4z)}$ V/m

$\Rightarrow E_0 = 11.2$ V/m

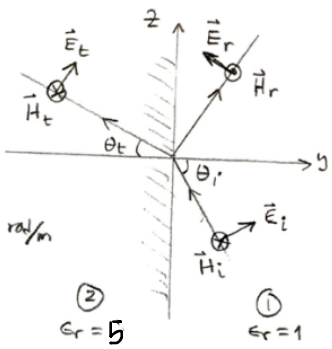
$\dfrac{\sin\theta_i}{\cos\theta_i} = \dfrac{10}{5} \Rightarrow \theta_i = 63.4°$

TM polarization, linear.

$\beta_1 = \sqrt{4+16} = \sqrt{20} \Rightarrow \beta_2 = \sqrt{100} = 10$ rad/m

$\lambda_1 = 2\pi/\beta_1 = 1.405$ m $\qquad \lambda_2 = 2\pi/\beta_2 = 0.628$ m

$\omega = \beta v = \beta_1 c = \sqrt{20} \times 3 \times 10^8 = 1.34 \times 10^9$ rad/s



$\theta_t = \sin^{-1}\left(\frac{1}{\sqrt{5}}\sin 63.4\right) = 23.57 \Rightarrow \Gamma_{TM} = \dfrac{\tan(-39.83)}{\tan(86.97)} = -0.044$

$\tau_{TM} = 0.467$

$\vec{E}_t = (11.2)(0.467)(0.4\vec{a}_y + 0.92\vec{a}_z)\,e^{j10(0.92y - 0.4z)}\ (V/m)$